

# Classification of

# ML Systems

**Supervised  
Learning**

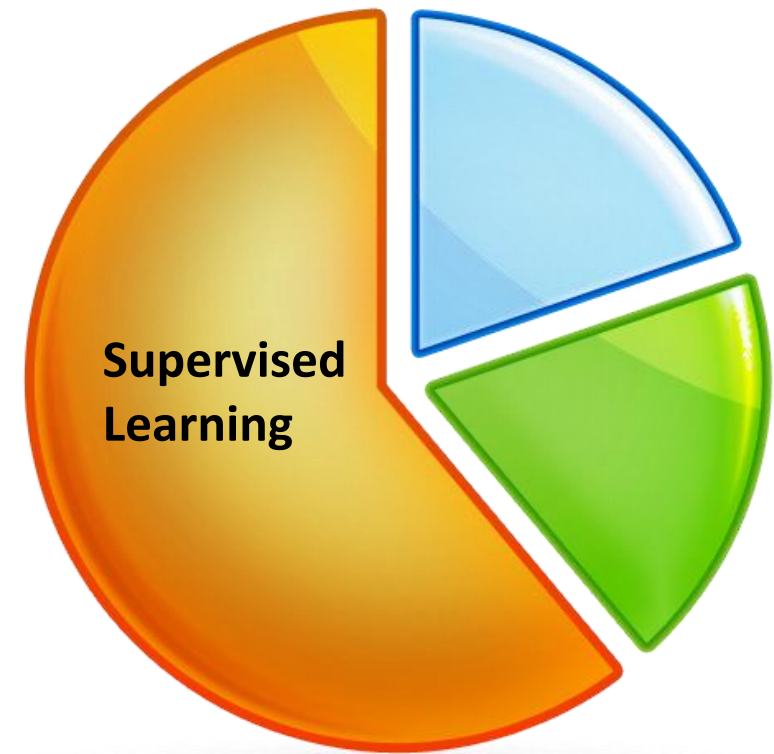
**Unsupervised  
Learning**

**Reinforcement  
Learning**

# #1 – Supervised Learning

## Overview

The most common  
**use case** of **ML**

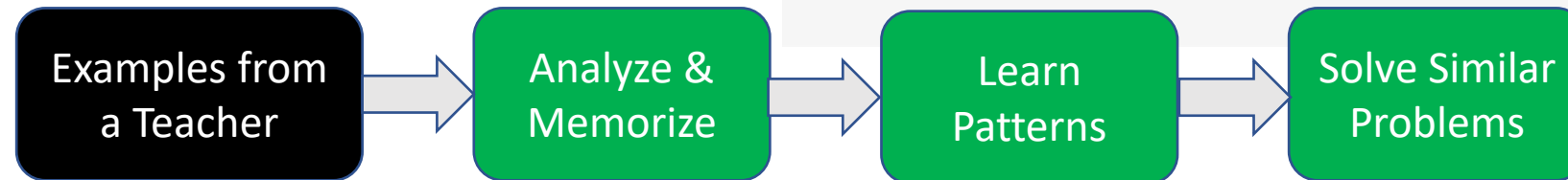


# #1 – Supervised Learning

## Overview

“Supervised”

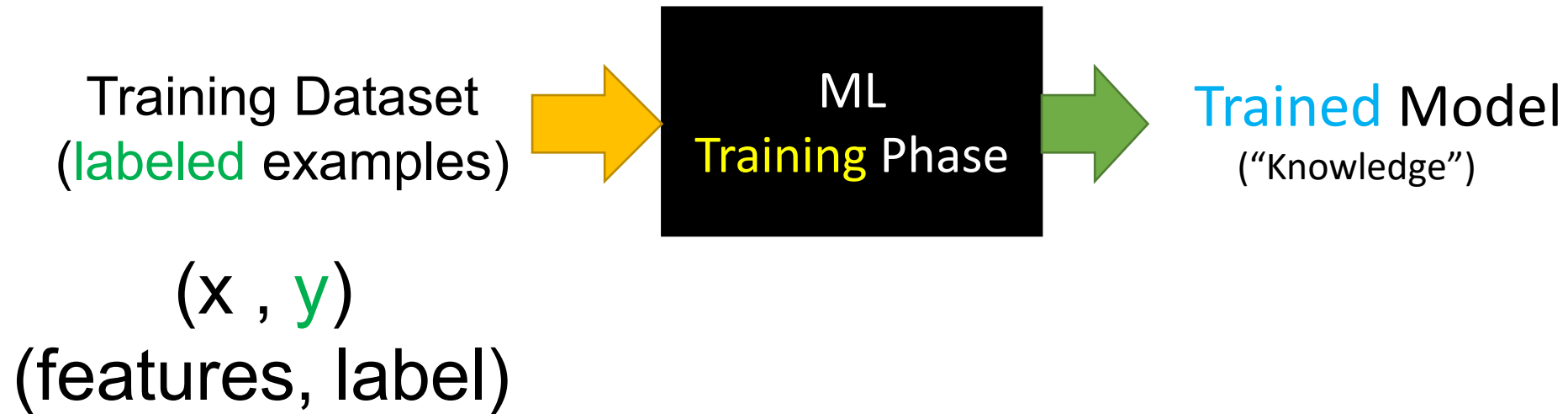
Learning under the supervision  
of a Teacher



Example 1  
Example 2  
Example 3

# #1 – Supervised Learning

## Overview



# #1 – Supervised Learning

## Overview

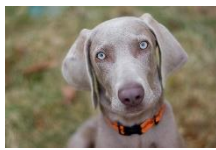
Training Dataset  
(labeled examples)



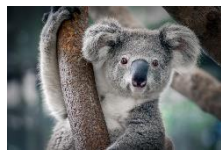
Dog



Not Dog



Dog



Not Dog

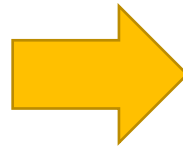


Not Dog



Dog

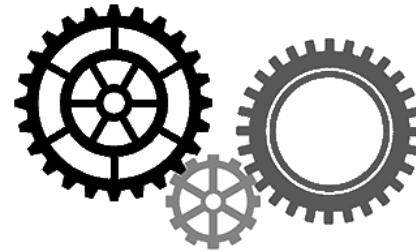
Cleaning the dataset



ML  
Training Phase

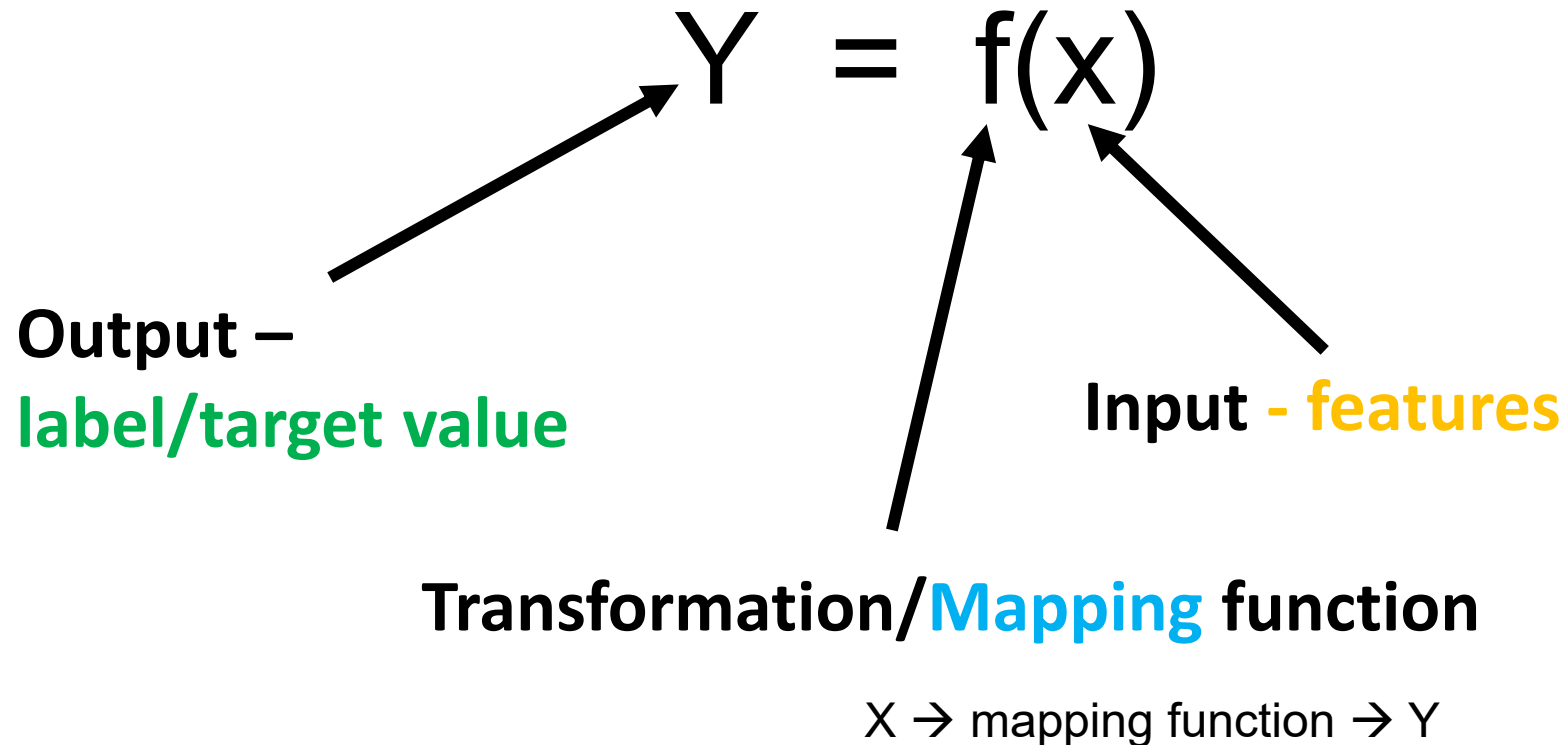


Trained  
Model



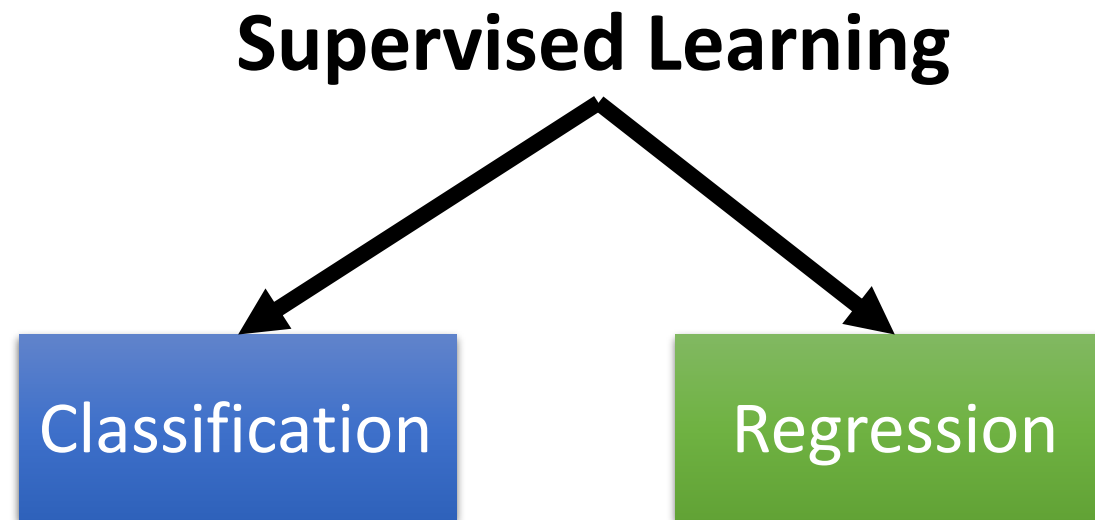
# #1 – Supervised Learning

## Mapping Function



# #1 – Supervised Learning

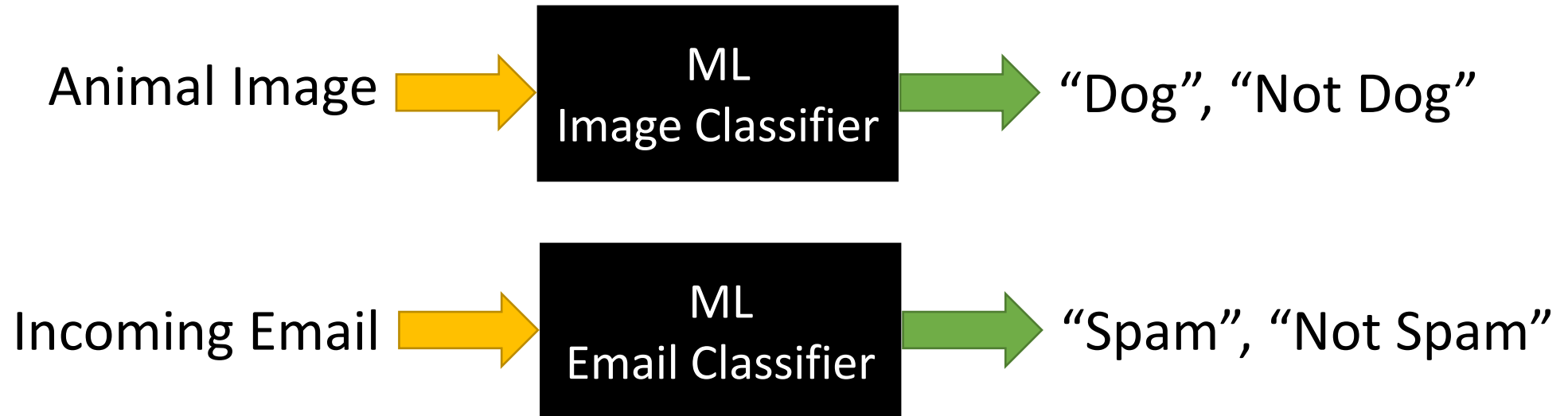
## Typical Tasks



# #1 – Supervised Learning

## Classification

### Binary Classification



# #1 – Supervised Learning

## Classification

### Multiclass Classification



Shallow Learning

Deep Learning



Support Vector Machines (SVMs)

# #1 – Supervised Learning

## Classification

Support Vector Machines (SVMs)

X1 - Height  
X2 - Weight



ML  
Gender Classifier



“Male”, “Female”

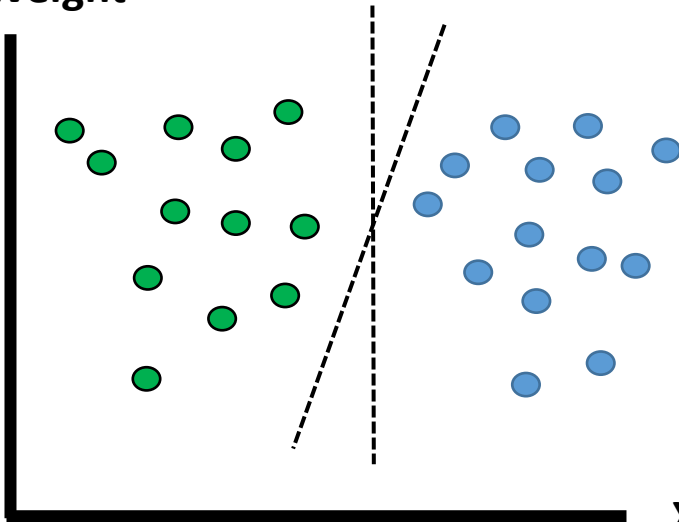
| X1 – H | X2 - W | Gender |
|--------|--------|--------|
| 160    | 56     | F      |
| 167    | 74     | M      |
| 183    | 85     | M      |
| .....  |        |        |

# #1 – Supervised Learning

## Classification

Support Vector Machines (SVMs)

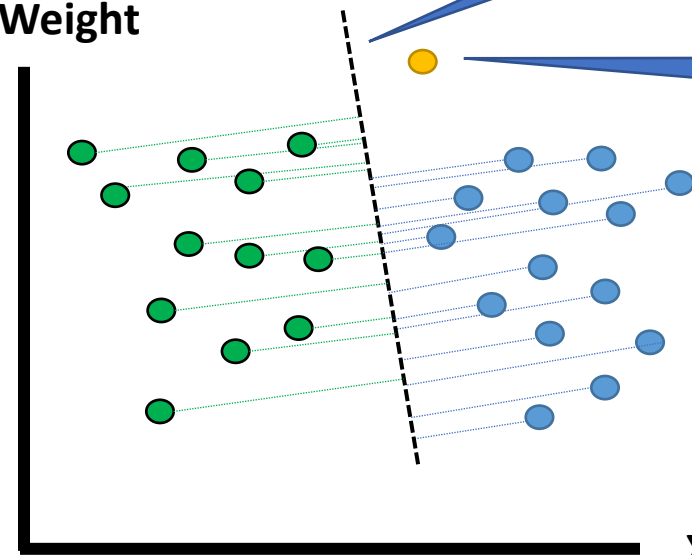
X2 - Weight



X1 - Height

Maximum Margin

X2 - Weight



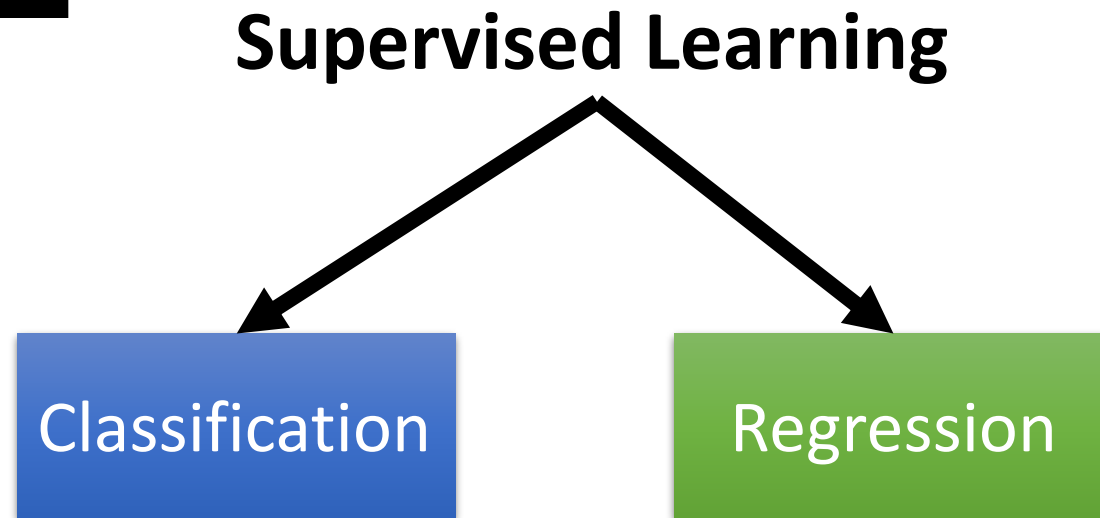
Optimal Line/Plane

This point is "Male"

X1 - Height

# #1 – Supervised Learning

## Classification



Support Vector Machines (SVMs)

# #1 – Supervised Learning

## Regression

Statistical method to analyze and predict data

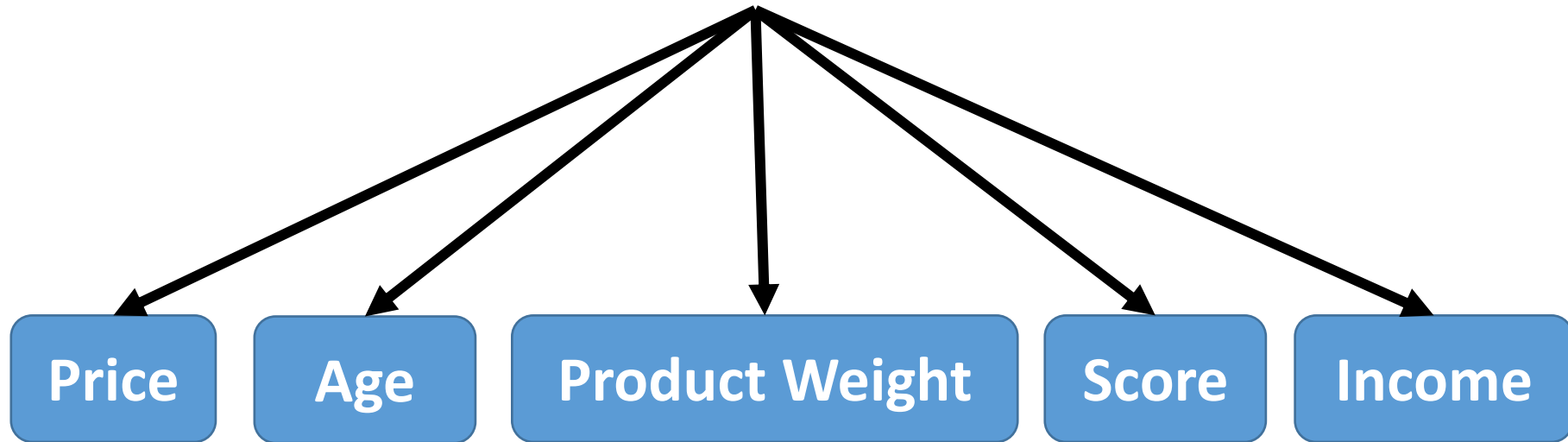
Predict a **continuous number**



# #1 – Supervised Learning

## Regression

Continuous Number



# #1 – Supervised Learning

## Regression

### What is Regression?

Statistical methods for estimating the **strength** of the relationship between a **dependent** variable and one or more **independent** variables.

Linear Regression

Logistic Regression

Polynomial Regression

# #1 – Supervised Learning

## Regression

### Linear Regression

$$Y = F(x_1, x_2, \dots, x_n)$$

**Dependent**  
(predicated)

**Independent**  
(predictors)

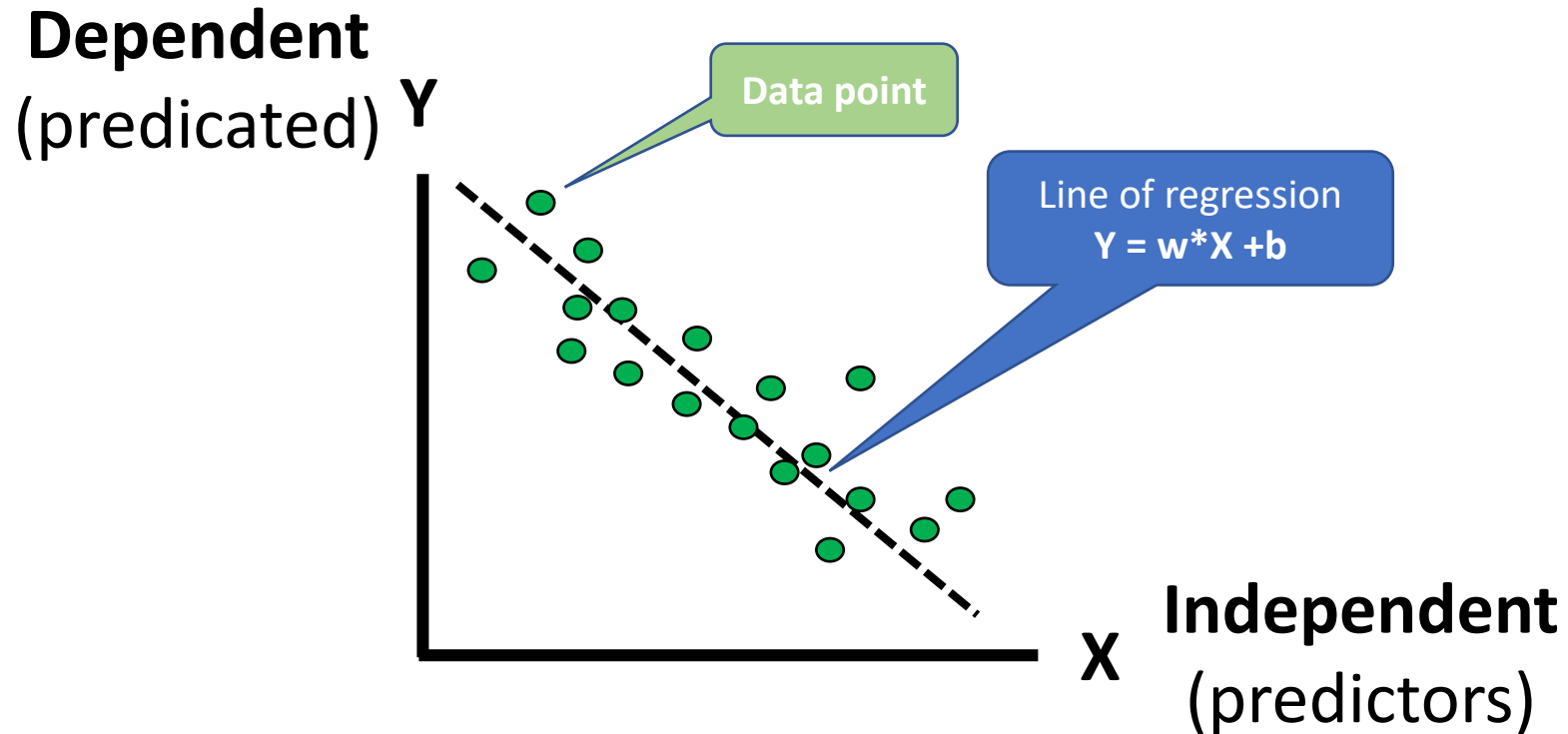
F - Linear Approximation

A black arrow points from the text 'F - Linear Approximation' up to the function 'F' in the equation  $Y = F(x_1, x_2, \dots, x_n)$ .

# #1 – Supervised Learning

## Regression

### Linear Regression



# #1 – Supervised Learning

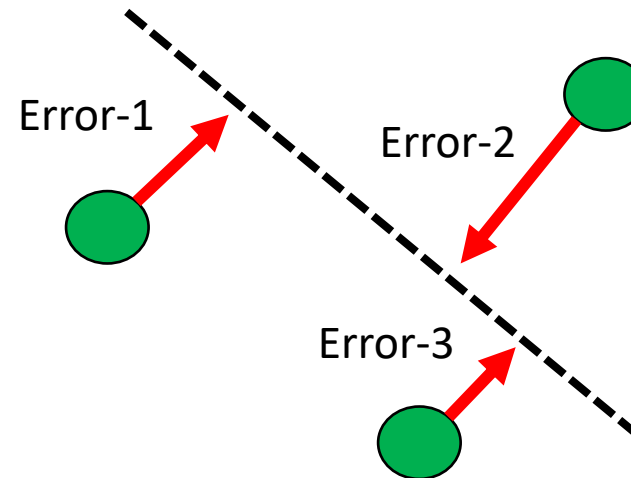
## Regression

### Linear Regression

Optimization using  
a **Cost** Function

Minimum(**MSE**)

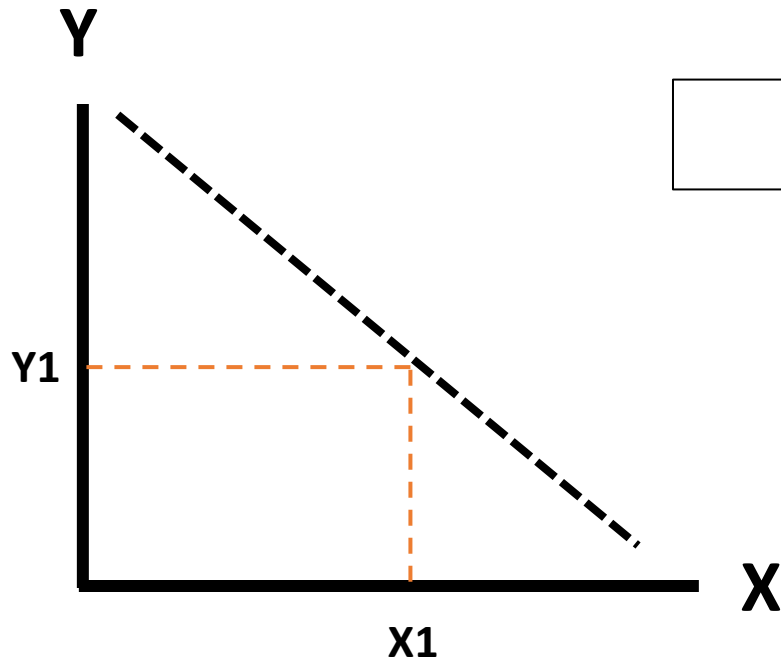
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$



# #1 – Supervised Learning

## Regression

### Linear Regression



$$y = w[0]*x[0] + w[1]*x[1] + \dots + w[i]*x[i] + b$$

$x[i]$  – input features

$w[i]$ ,  $b$  – model parameters

# #1 – Supervised Learning

## Summary

### Supervised Learning

```
graph TD; A[Supervised Learning] --> B[Classification]; A --> C[Regression];
```

Classification

Support Vector Machines (SVMs)

Regression

Linear Regression  
Logistic Regression  
Polynomial Regression

# Supervised Learning



The vast majority of available data in many applications is usually unlabeled

Buy a labeled  
dataset

Manually label a  
dataset

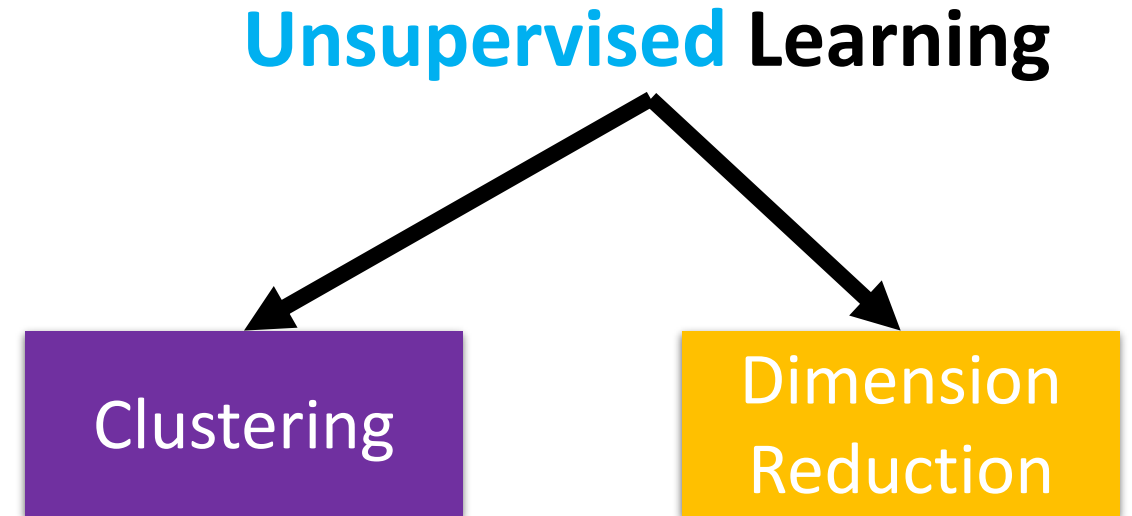
Unsupervised  
learning

# #2 – Unsupervised Learning

## Overview

Learning **without** a “teacher” supervising the learning process

Identify **automatically** meaningful patterns in unlabeled data



# #2 – Unsupervised Learning

## Clustering

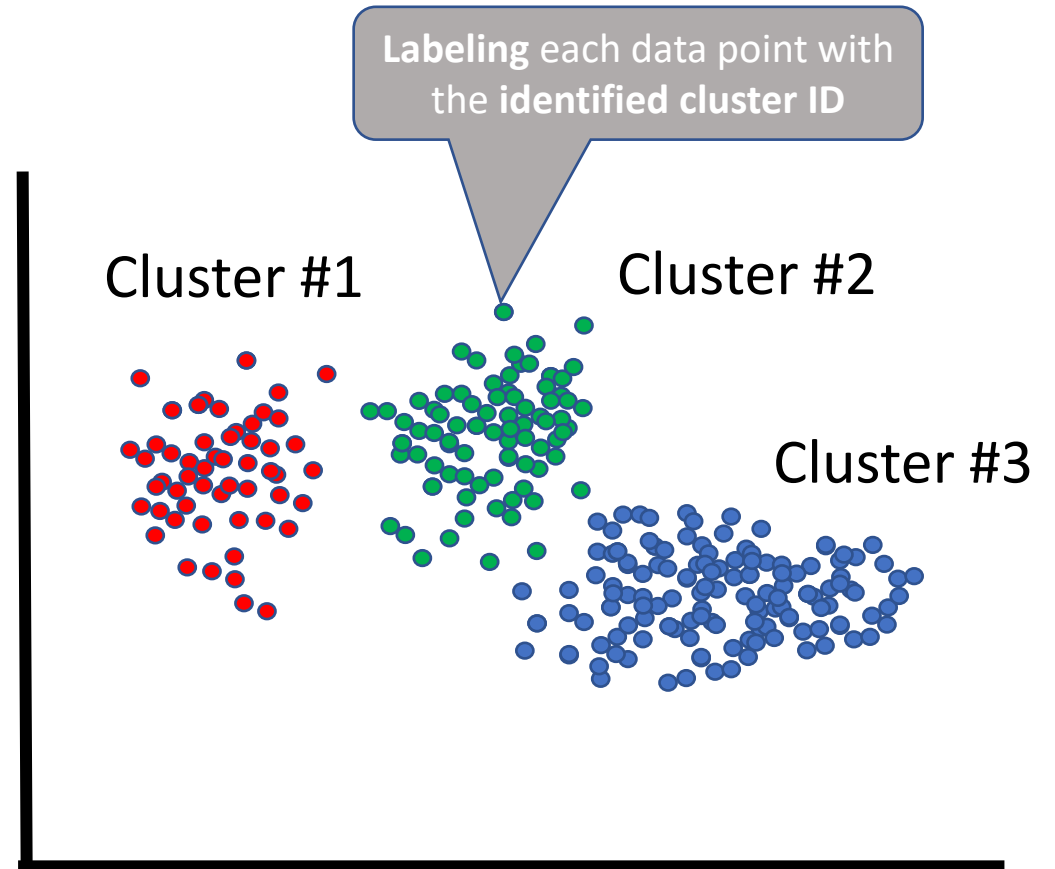
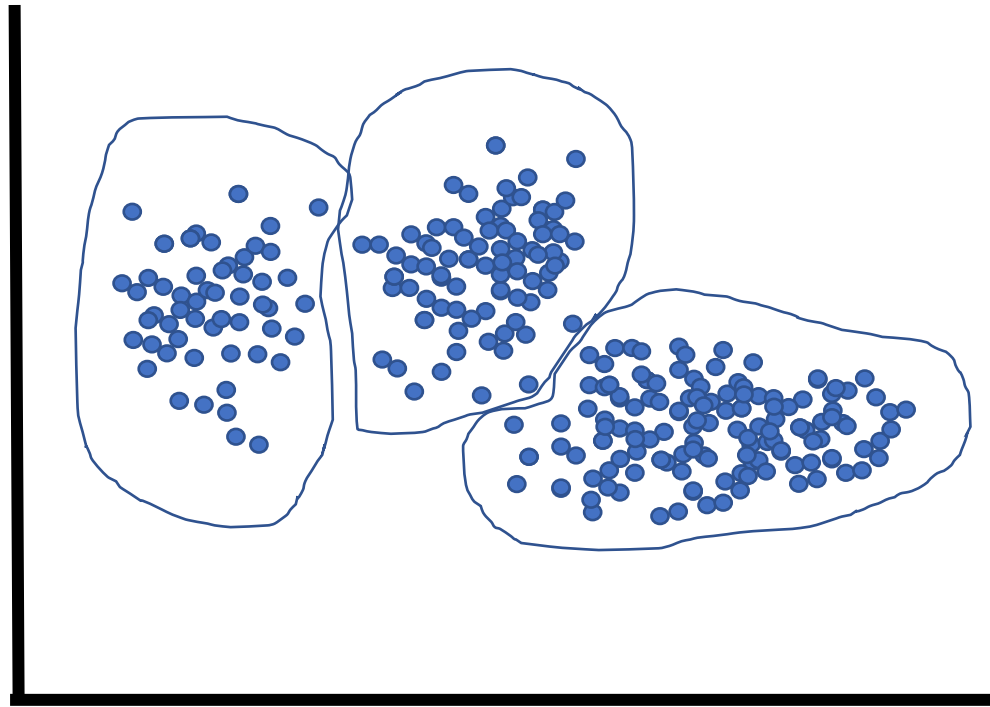
### What is Clustering?

Clustering is the task of identifying similar instances with **shared attributes** in a dataset and **group them together into clusters**

The **output** of the algorithm would be a set of “**labels**” assigning each data point to one of the identified clusters

# #2 – Unsupervised Learning

## Clustering



# #2 – Unsupervised Learning

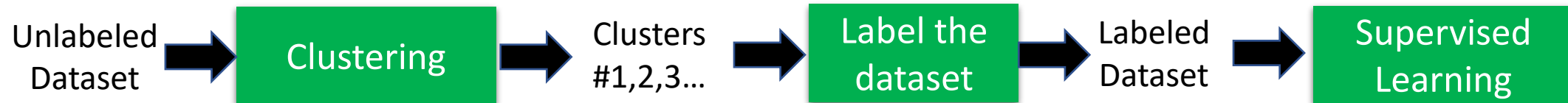
## Clustering

### Use Cases for Clustering

Customer Segmentation

Anomaly/Outlier Detection

Semi-supervised Learning



# #2 – Unsupervised Learning

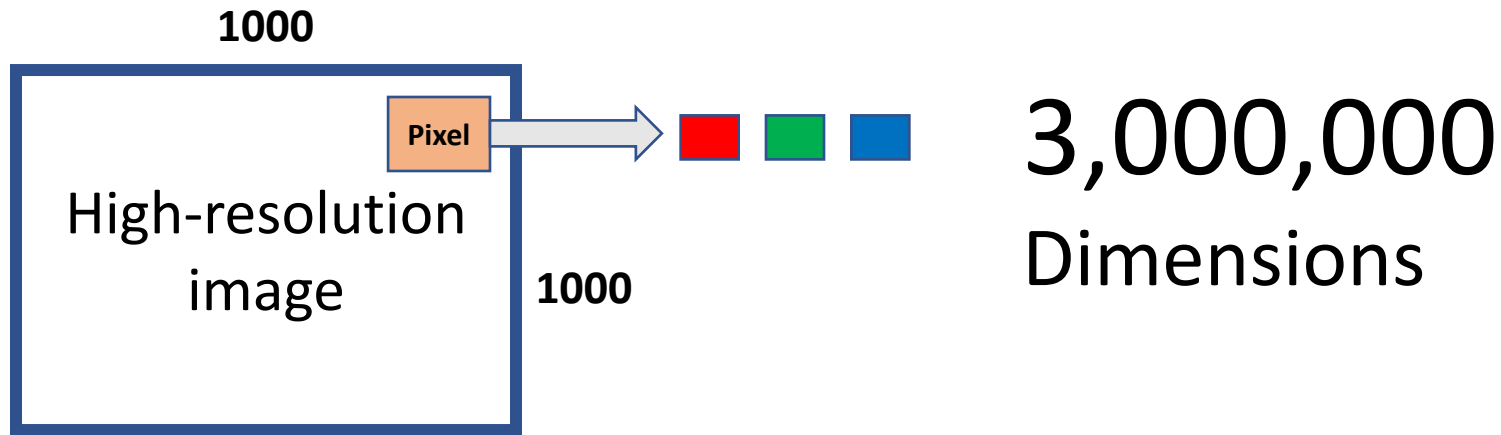
## Dimension Reduction

In supervised learning, one **BIG** challenge to handle is the **number of input features** that the algorithm needs to analyze



# #2 – Unsupervised Learning

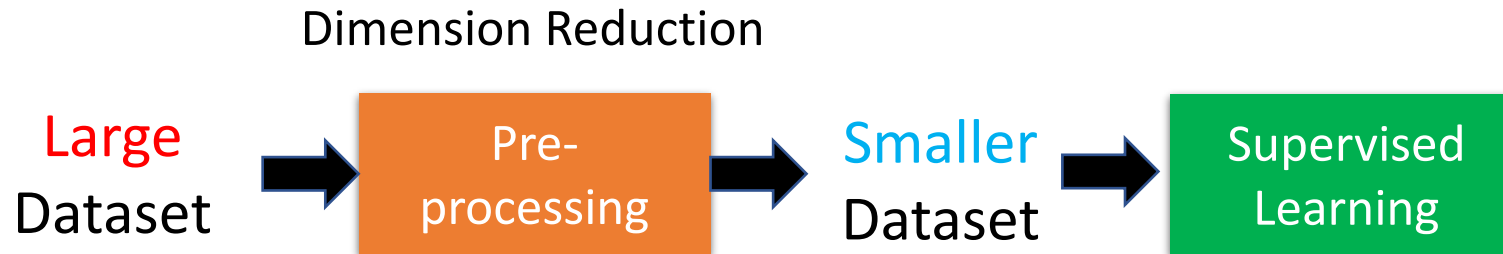
## Dimension Reduction



In many cases, not **ALL** features are needed...

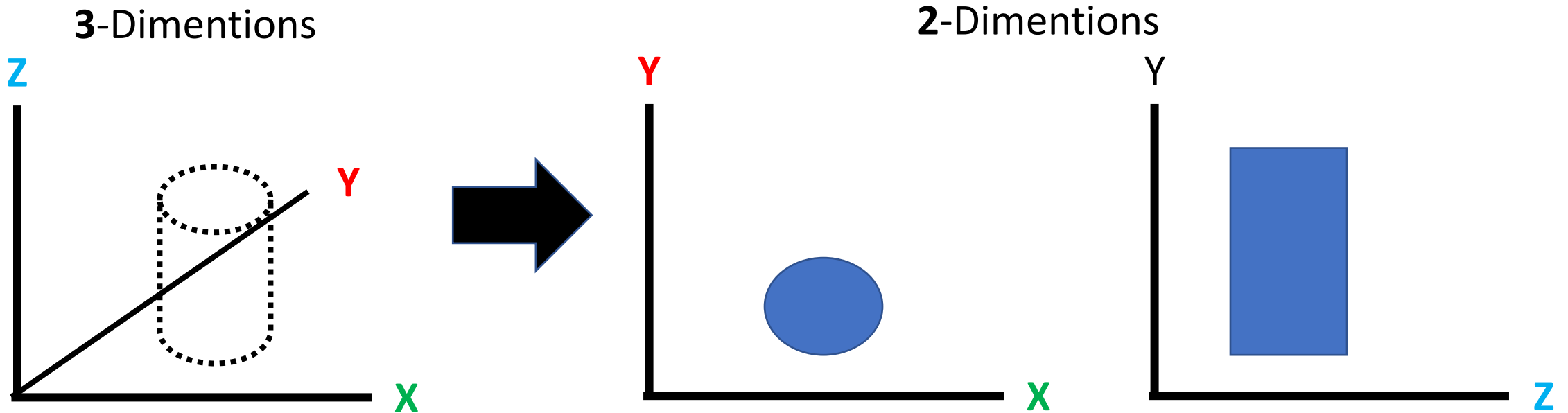
# #2 – Unsupervised Learning

## Dimension Reduction



# #2 – Unsupervised Learning

## Dimension Reduction



# #2 – Unsupervised Learning

## Dimension Reduction



10,000 Images

640x640 resolution

Pixel (RGB) –  $256 \times 256 \times 256 = 16.7\text{M}$  possible colors

1 Image =  $640 \times 640 \times 3 = 1.2\text{M}$ bytes

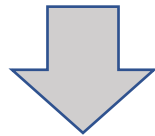
# #2 – Unsupervised Learning

## Dimension Reduction

Pixel (RGB) :  $256 \times 256 \times 256 = 16.7\text{M}$  possible colors



Pixel : 256 possible colors



1 Image =  $640 \times 640 \times 1 = 0.4\text{M}$  bytes

# #2 – Unsupervised Learning

## Dimension Reduction

Clustering pixels according to their color then replacing each pixel's color with the mean color of its cluster



# #3 – Reinforcement Learning

## Overview

We are **not using** a group of labeled or unlabeled examples

Used as a framework for **decision-making tasks**  
based on goals

Perform a complex objective while performing multiple  
**sequences of actions**

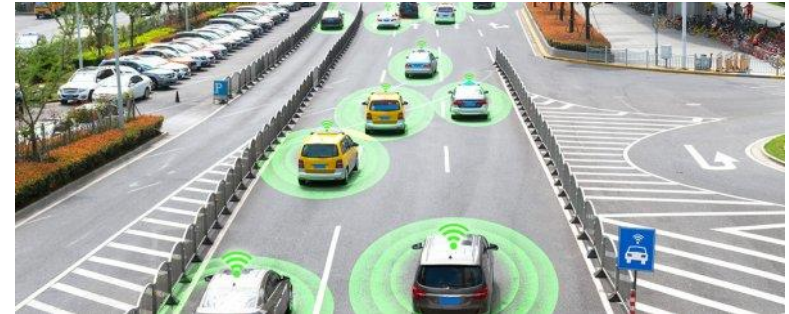
# #3 – Reinforcement Learning

## Overview

### Playing Computer Games



### Handling Dynamic Environment



# #3 – Reinforcement Learning

## Overview

Win the game by deciding how to play **multiple turns**

Every move has thousands of **future options** to consider

Every move is changing the **ongoing state** of the game

**It is a dynamic environment!**

The feedback on selected strategy is **delayed**

**Interaction** between a machine and the environment (other player)

**Reinforcement Learning**



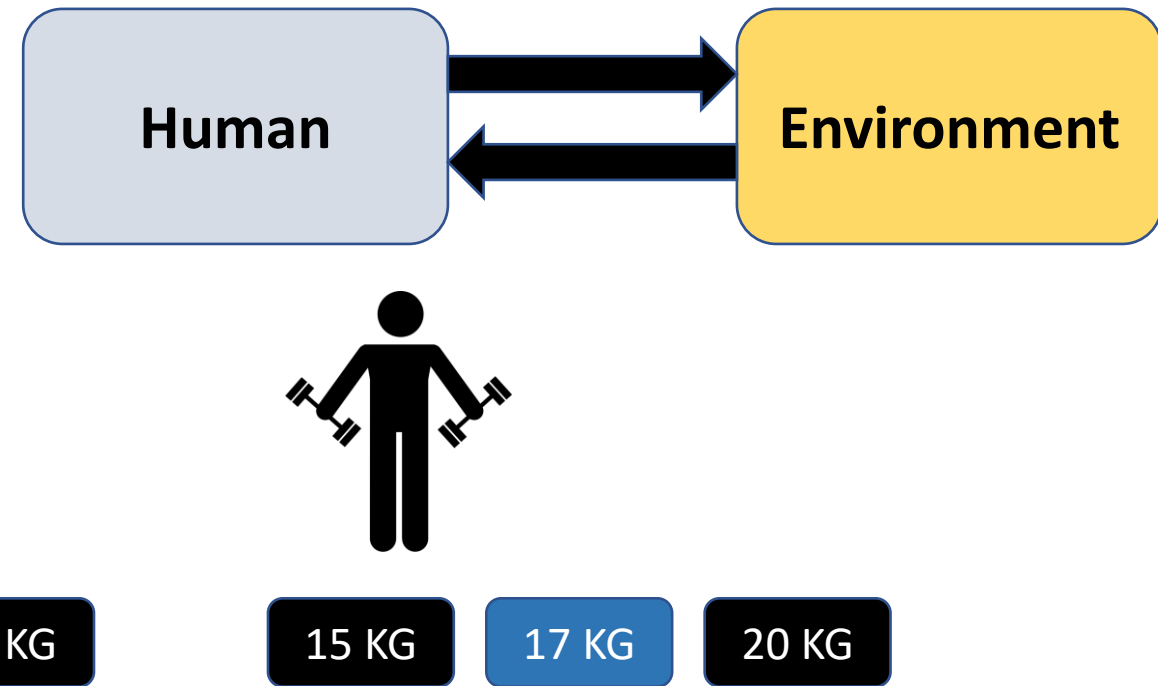
# #3 – Reinforcement Learning

## Learning from Interaction

The concept of reinforcement learning is very similar to the way **humans** are learning

We **try things** and get **feedback**, and based on the feedback we are learning

Learning from the **experience**



# #3 – Reinforcement Learning

## Learning from Interaction

Interaction with the car

Interaction with the road conditions

interaction with other drivers

Learning from the  
interaction is a foundational  
idea in our daily life



Reinforcement Learning

# #3 – Reinforcement Learning

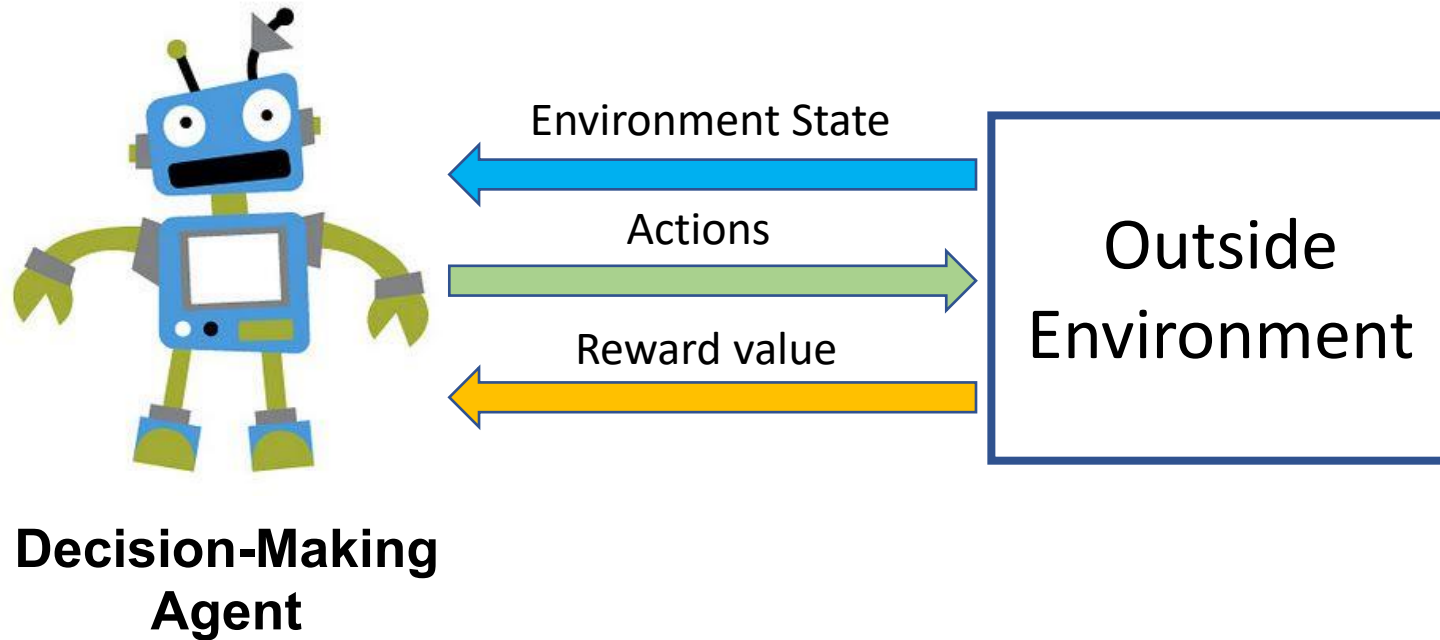
## The RL Concept

Reinforcement learning is a method being used to let machines learn how to behave based on **interaction** with the environment while focusing on some **end goal**



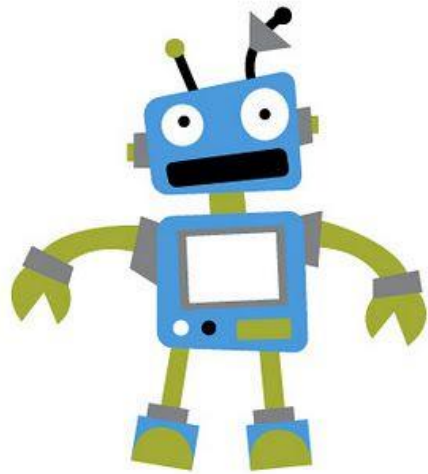
# #3 – Reinforcement Learning

## The RL Concept

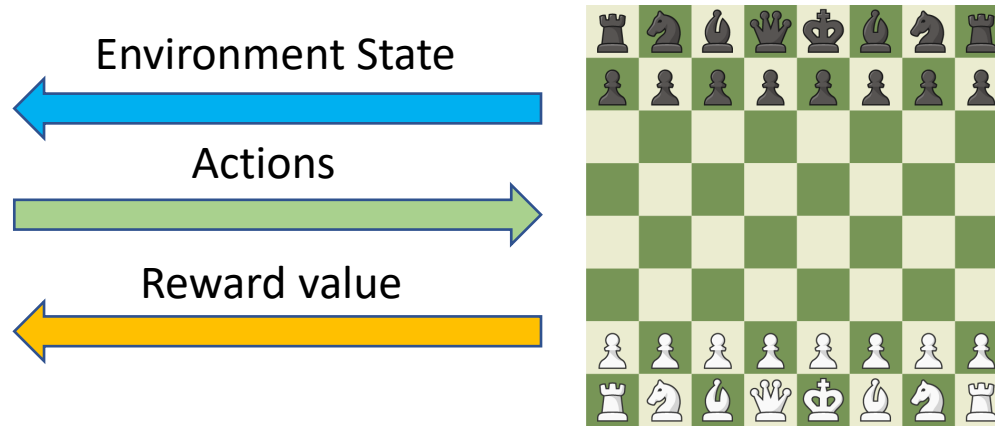


# #3 – Reinforcement Learning

## The RL Concept



Chess-playing Agent



# #3 – Reinforcement Learning

## The RL Concept

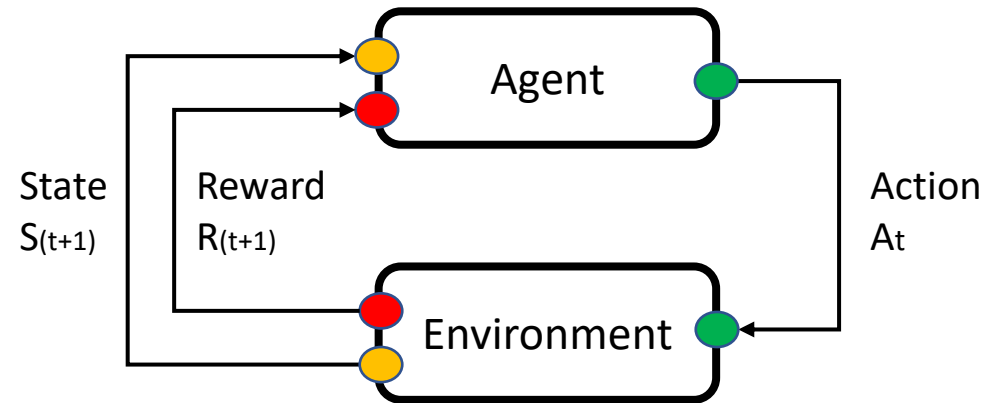
Multiple steps on a time dimension

$t = 1, 2, 3, 4, \dots$

Step  $t$ :



$[S_0, A_0, R_1], [S_1, A_1, R_2], [S_2, A_2, R_3], \dots$



# #3 – Reinforcement Learning

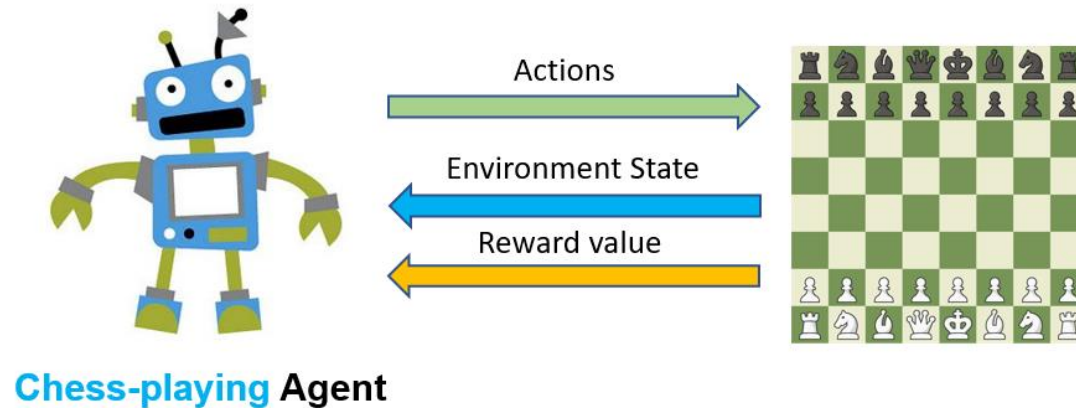
## Trial-and-Error Search



Reinforcement learning builds a prediction model by gaining feedback from **random trial and error** and leveraging insight from previous iterations

# #3 – Reinforcement Learning

## Trial-and-Error Search



Losing ☹️



Winning 😊

# #3 – Reinforcement Learning

## Summary

Reinforcement learning is used in applications that the machine must make a **sequence of decisions**, and those decisions are coming with **positive** or **negative** consequences

The cumulative knowledge of how to achieve a specific goal is **reinforced** again and again by experience

